***Note:*** Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. Students should take the signature from the faculty in charge. For every laboratory session, the observation component carries ***marks***.

## *What is Python?*

Python is an interpreted, high-level programming language emphasizing readability. It is suitable for Rapid Application Development, scripting, and connecting components due to its dynamic nature and data structures.

Uses

- Web and Internet Development
- Scientific and Numeric
- Education
- Desktop GUIs
- Software Development
- Business Applications

***Various Ways to Execute Python***

## *1.Terminal Execution:*

- Open Notepad, write your Python code, and save the file with a .py extension (e.g., script.py) by selecting "All Files" in the "Save as type" option.
- Open the Terminal/Command Prompt, navigate to the folder where your Python file is saved using the cd command.
- Run the Python file by typing python script.py (or python3 script.py depending on your system), and the code will execute in the terminal.

```
print("hello, this is python")
```

```
Command Prompt
Microsoft Windows [Version 10.0.19045.4780]
(c) Microsoft Corporation. All rights reserve
d.

C:\Users\ChitraGM>cd Desktop

C:\Users\ChitraGM\Desktop>python 1.py
hello, this is python

C:\Users\ChitraGM\Desktop>
```

### *2.IDLE (Integrated Development and Learning Environment) :*

- Itserves as a convenient toolfor writing,testing, and debuggingPython code.
- Python IDLE provides a graphical user interface (GUI) that simplifies the coding process and helps users manage their Python projects efficiently.
- Python's IDLE capability enables programmers to write and execute a single line of code, much as how Shell writes and edits code.
- Itisextremely user-friendly, especially for novices, due to features like intelligent indentation and highlighting.

 ***Install IDLE***

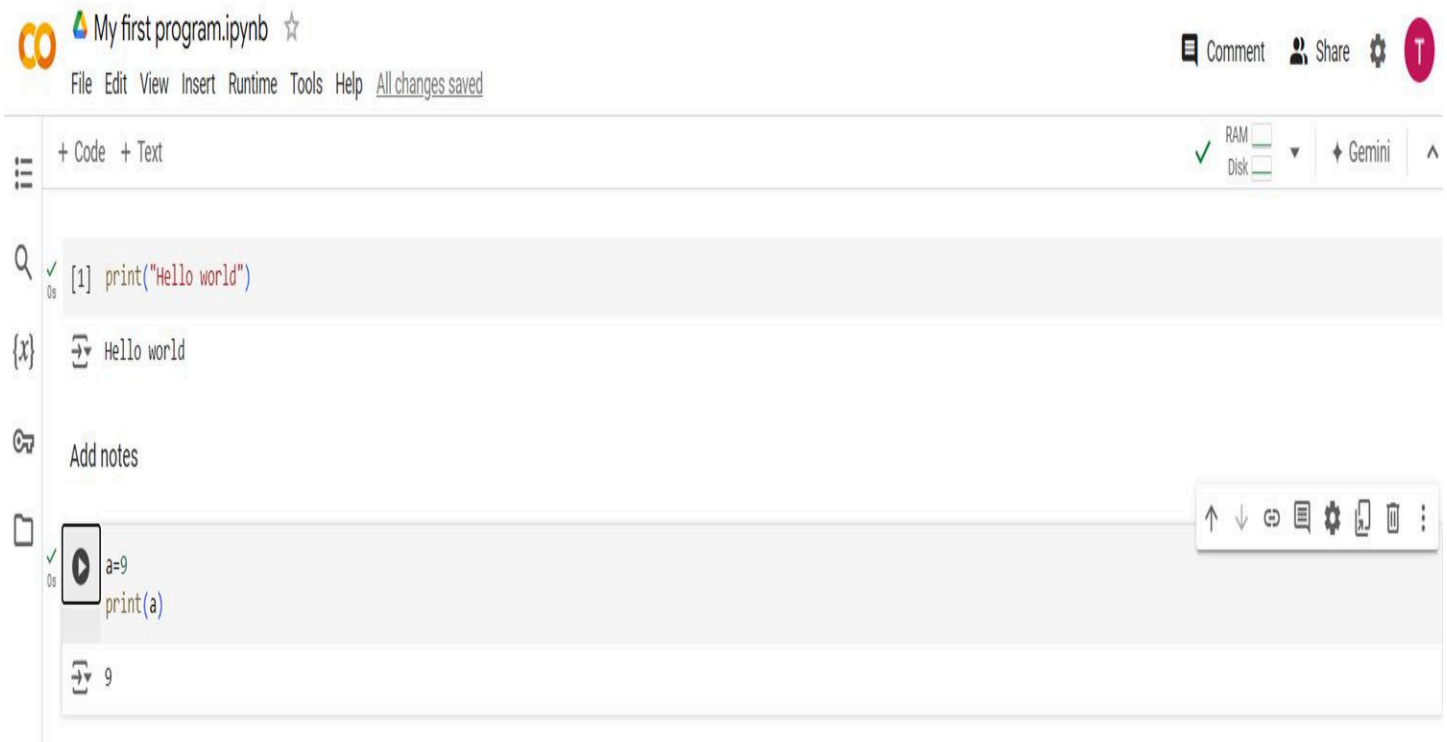
- ***Link for Linux***
- ***Link for Windows***
- ***Link for Mac***

HereareYoutubelinksfor easy installation for the different Oses.

### *3.Google Colab:(https://colab.research.google.com/)*

Colab,or"Colaboratory",allows youtowriteandexecutePythoninyour browser, with:

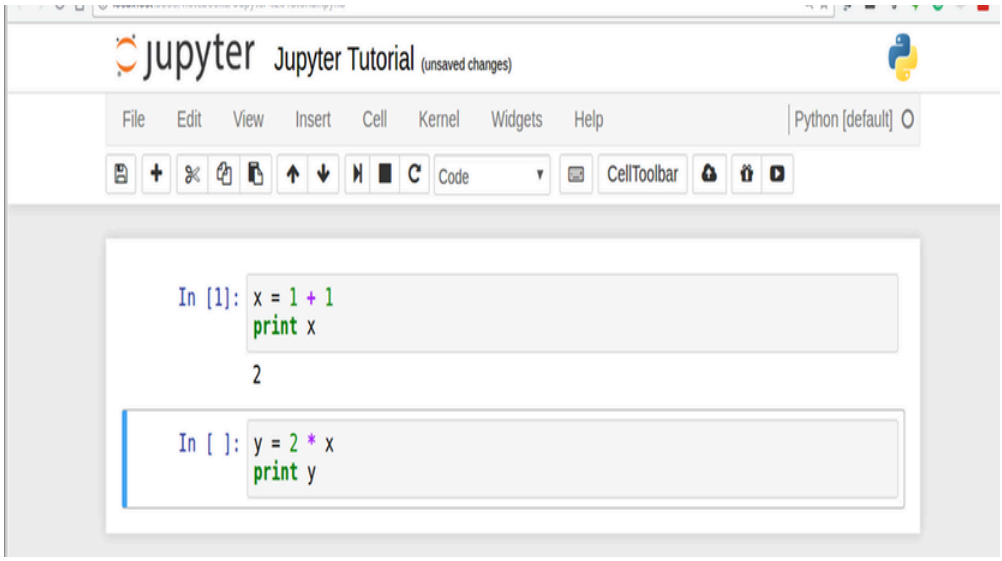
- Zero configuration required
- Access to GPUs free of charge
- Easy sharing



## *4.Jupyter Notebook:*

Jupyter Notebook isan interactive web-based tool that allows users to write and run code in real-time, making it ideal for learning Python. Key features include

- Cell-based execution
- Easy debugging
- Support for data visualization libraries like Matplotlib.




 ***Install Jupyter Notebook***

- ***Link for Linux***
- ***Link for Windows***
- ***Link for Mac***

Hereare Youtube links for the different Oses.

## *4. VS Code :*

VisualStudioCode (VS Code) is a lightweight, open-source code editor by Microsoft, supporting multiple programming languages like Python, JavaScript, and C++. Its fast interface is favored by developers for coding, debugging, and collaboration. Key features include:

- Extensive extensions for languages, themes, and tools
- Code suggestions and automatic completion
- Built-in debugger for identifying and fixing code errors

***Install VS Code:***

- ***Link for Linux***
- ***Link for Windows***
- ***Link for Mac***

Here are Youtube links for the different Oses.

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